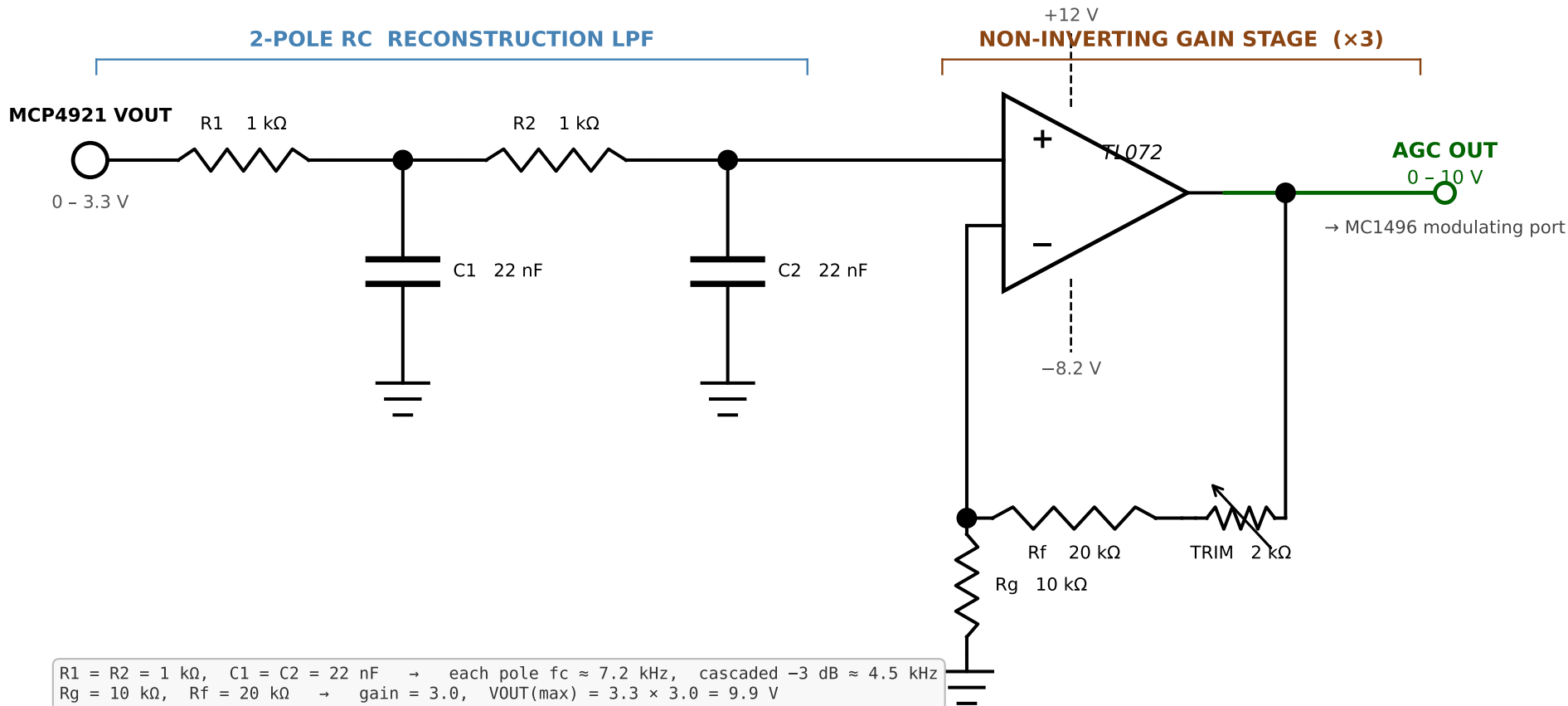


20 m CW TX — Key Shaping Chain: MCP4921 DAC → LPF → Op-Amp → MC1496



R1 = R2 = 1 kΩ, C1 = C2 = 22 nF → each pole $f_c \approx 7.2$ kHz, cascaded -3 dB ≈ 4.5 kHz
Rg = 10 kΩ, Rf = 20 kΩ → gain = 3.0, VOUT(max) = 3.3 × 3.0 = 9.9 V
Adjust TRIM to calibrate VOUT(max) = 10.0 V exactly for full MC1496 AGC range.
Op-amp supply: use rig +12 V / -8.2 V rails (already present).